TraceErase – Our Solution Explained

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Our solution

TraceErase

AI-Powered Personal Data Deletion and Verification Platform

Find it. Delete it. Prove it.

TraceErase finds every company that holds your personal data, gets it deleted from those companies **and** from every partner they secretly passed it to, and then **proves** whether the deletion really happened, with evidence nobody can fake or alter.

Team	Cranial Remission
Event	DEXATHON 2026
Track	Blockchain and Cybersecurity
Problem statement	Problem 4: Verifying whether personal data has actually been deleted
Who it is for	Every person who has signed up for apps and websites (and, through public scores, the companies and regulators too)

The name. *Trace* is what we do first: trace where your data went, including the companies you never dealt with directly. *Erase* is what we make happen: deletion, everywhere. Together the name says the whole idea: you cannot erase what you cannot trace, and you cannot trust an erase you cannot check.

The problem

When you ask a company to delete your data, you have no way to check that it actually did, or that the partners it shared your data with did too. You get, at best, a one-line email saying "your account has been deleted", and you are asked to simply trust it.

1. Our personal data is spread across dozens of companies

Every app and website we sign up for collects something: name, email, phone number, home address, payment details, location, order history, sometimes ID documents. Over the years most people collect accounts they have forgotten about: a food app used once, a travel site from a holiday, a shopping site from a sale. Nobody keeps a list. So the first problem is simple: **people do not even know who holds their data.**

2. Companies pass our data on to others we never met

A food delivery app does not keep your data to itself. It shares it with a payment company to take your money, delivery partners to bring the food, analytics tools to study the app, and often advertising networks to show you ads. These are real companies holding your email or phone number, and **you never signed up with any of them.** You usually don't know their names.

3. Asking for deletion is slow and confusing

Every company has its own way: an email to a grievance officer whose address is buried in a long privacy policy, a web form, a setting deep inside the app, or a support chat. The request has to be written properly, quoting the right law. Many people start, get confused and give up. Doing it for 40 companies by hand is not realistic.

4. Even when a company says "deleted", nobody can check it

This is the heart of our problem statement. The company's reply is just words. It may have:

- deleted the account but kept the data in backups or other systems;
- kept some data for a legal reason without telling you which data or why;
- deleted its own copy but **never told its partners**, who still have your data;
- or simply ignored you.

The person cannot see inside the company's servers, so today there is no proof either way.

5. Leaks happen, and you can't tell who leaked

You start getting spam emails, loan calls and promotional SMS from companies you never gave your details to. Someone passed your data on, or sold it. But because you used the same email and number everywhere, **it is impossible to tell which company was the source.** Without knowing the source, you cannot complain about it.

6. Ignored requests leave you with no evidence

If a company ignores your request, the law lets you complain. But a complaint needs proof: when you asked, what you asked, that you reminded them, that they did not answer. Most people have scattered emails, no dates, and no proof of what was sent. Their complaint is weak before it starts.

7. The law gives us the right, but not the tool

India's **Digital Personal Data Protection Act, 2023 (DPDP Act)** gives every person (called a *Data Principal*) the right to have their personal data erased, and requires companies (called *Data Fiduciaries*) to erase data when consent is withdrawn and to make their partners (*Data Processors*) erase it too. The **DPDP Rules, 2025**, notified in November 2025, give companies up to **90 days** to respond to such requests, set up a fully digital **Data Protection Board**, and come into full effect in phases ending around **May 2027**. Penalties go up to **₹250 crore**. Similar rights exist worldwide: the GDPR's "right to erasure" in Europe and the "right to delete" in California.

But no law gives the person a way to **verify** that deletion happened. The right exists on paper; checking it is left to trust.

The problem in one line

> People cannot find who has their data, cannot easily get it deleted everywhere, and cannot prove whether it was ever really deleted.

Why it is needed today

This problem has always existed, but four things make it urgent **now**: the right to erasure is becoming enforceable in India, data leaks keep growing, spam and scam calls have become a daily nuisance for almost everyone, and nobody yet offers a way to check that companies actually comply.

1. The law is switching on right now

The DPDP Rules were notified in November 2025 with an 18-month phased rollout. Registration of *consent managers* begins around November 2026, and the remaining provisions, including **data erasure**, security safeguards and breach reporting, apply by about **May 2027** (PIB, Legal 500).

That means hundreds of millions of Indians are about to gain a right that most will want to use, and thousands of companies will start replying "done". The moment a right becomes enforceable, people need a way to **check** it was respected, and the Data Protection Board

needs complaints that come with **evidence**. TraceErase is built for exactly this moment.

2. Our data keeps leaking, at enormous scale

Large Indian breaches have exposed the details of hundreds of millions of people (Corbado summary of reported breaches):

Organisation	Year	People affected (reported)	Data exposed
ICMR (COVID-19 test data)	2023	About 815 million	Names, Aadhaar and passport details, phone numbers, addresses
Hathway (internet provider)	2024	About 41.5 million	Names, emails, phones, addresses, account details
boAt (electronics brand)	2024	About 7.5 million	Names, addresses, phones, emails, purchase history

The same summary reports that CERT-In handled about 1.32 million cybersecurity incidents between January and October 2023. Every extra company holding your data is one more place it can leak from. **Data that has been truly deleted cannot leak.** That makes deletion a security measure, not just a privacy wish.

3. Spam and scam calls have become daily life

A LocalCircles survey reported in September 2024 found that **95% of mobile users receive unwanted calls every day**, **77% get three or more a day**, and **96% of people on the Do Not Disturb list still get them**, mostly from financial services and real estate (Business Today). These callers got the numbers from somewhere. Blocking them one by one never finds the source; TraceErase's protected identity does.

4. More apps, more sharing, and still no way to check

Everyday life now runs on apps: food, groceries, payments, travel, shopping, gaming. Each one shares data with payment, delivery, analytics and advertising partners. Yet even as laws improve, there is still no tool that tells a person, with evidence, whether a company and its partners really deleted their data. Without that check, the right to erasure stays a promise on paper. **TraceErase turns it into something people can verify.**

Existing solutions and their drawbacks

Several tools touch parts of this problem, but none of them answers the real question: **did my data actually get deleted, everywhere it went?** Each one either finds, or asks, or hides, and then stops at trusting the company's word.

1. Doing it yourself

You find each company, dig out its grievance email from the privacy policy, write a legal request, track replies and deadlines, and send reminders.

- **Drawbacks:** takes hours per company; most people don't know who holds their data; partners are never reached; the only "proof" is the company's reply; if a company ignores you, you have scattered emails, not evidence.

2. Data-removal services (DeleteMe, Incogni, Optery and similar)

Paid subscriptions that send opt-out and deletion requests to **data brokers**: companies that collect and sell personal profiles. Incogni uses automated requests to hundreds of brokers; DeleteMe uses human specialists and sends reports with screenshots of removed listings (CyberInsider comparison).

- **Drawbacks:** they target data brokers, not the apps you actually signed up for (Zomato, Swiggy, Amazon) or those apps' partners; coverage is mostly the US, Europe, UK, Canada and a few other countries, not India; they cost roughly US$95 to US$129 a year; "proof" is a status dashboard or a screenshot that a public listing disappeared, not evidence that the data stopped being used; they do not trace leaks or prepare complaints.

3. Request-sending apps (Mine, Permission Slip)

- **Mine** scans the subject lines of your Gmail or Outlook inbox to find companies that have your data, then generates deletion requests with a "reclaim" button (VentureBeat). It now also sells privacy software to businesses (Mine).
- **Permission Slip** (Consumer Reports) acts as your *authorized agent* under US state privacy laws and sends requests to companies for you, then shows a status when a request is fulfilled or rejected (Consumer Reports).
- **Drawbacks:** they **send** requests but do not **verify** deletion; "completed" just means the company said so; they do not follow your data to partners; they do not detect or trace leaks; no tamper-proof evidence for a regulator; built around GDPR and US laws rather than India's DPDP Act.

4. Email and phone masking (Apple Hide My Email, Firefox Relay, DuckDuckGo Email Protection)

They give you alias email addresses that forward to your real inbox, so sites never see your main address. Deleting an alias stops mail to it (Nerdschalk).

- **Drawbacks:** they only **hide**; they never ask a company to delete anything; you might notice which alias gets spam, but nothing investigates who leaked it, records evidence or acts on it; aliases are not linked to deletion, so silence on an alias proves nothing to anyone; some need a paid subscription and are tied to one ecosystem.

5. Spam blockers and Do Not Disturb (caller ID apps, TRAI's DND registry)

They label or block unwanted calls and SMS after they arrive.

- **Drawbacks:** they treat the symptom, not the cause; they never tell you **which company** leaked your number; nothing gets deleted; and the survey above shows 96% of people on DND still get unwanted calls.

6. Company-side privacy software and consent managers

Companies buy software to manage incoming privacy requests, and under the DPDP Rules, registered **consent managers** will help people give and withdraw consent from about November 2026.

- **Drawbacks:** company tools serve the **company**, so the same organisation that holds your data also reports whether it deleted it; consent managers manage permissions going forward but do not check whether data already shared was erased, or follow it to partners.

Side by side

	Find who has your data	Send deletion requests	Reach the partners	Verify deletion	Trace leaks to the source	Evidence for a complaint	Built for India's DPDP Act
Doing it yourself	Hard	Yes, by hand	No	No	No	Weak	Yes
Data-removal services	Data brokers only	Yes, to brokers	No	Screenshots of listings	No	No	No
Request-sending apps	Yes (inbox scan)	Yes	No	No	No	No	No

	Find who has your data	Send deletion requests	Reach the partners	Verify deletion	Trace leaks to the source	Evidence for a complaint	Built for India's DPDP Act
Email masking	No	No	No	No	Partly (you may notice)	No	Not applicable
Spam blockers / DND	No	No	No	No	No	No	Partly
Company tools / consent managers	No	No	No	Company reports on itself	No	No	Yes
TraceErase	**Yes (AI)**	**Yes (AI letters, signed)**	**Yes, automatically**	**Yes: signed reply + 30-day silence**	**Yes (AI investigator)**	**Yes (blockchain-anchored)**	**Yes**

What makes the gap: every existing tool stops when the company says "done". TraceErase starts there. It is the only approach that combines a unique identity per company, AI that reads replies and follows the data to partners, silence-based verification, leak tracing, and tamper-proof evidence.

The solution

TraceErase is a personal privacy assistant, powered by AI, that does three jobs for you: it **finds** who has your data, **deletes** it everywhere including partners, and **proves** what really happened, with records that cannot be changed. It works with every company exactly as they are today; no company has to install, join or agree to anything.

The three promises

Promise	What TraceErase does	What you get
Find it	AI scans who has emailed you and where you signed in, then reads each company's privacy policy	A complete list of companies holding your data, what each one keeps, and who it shares with
Delete it	AI writes a legally correct deletion request for each company, you sign it once, TraceErase sends it, follows up and escalates	Deletion requests to every company, and to every partner they name, without you writing a single letter

Promise	What TraceErase does	What you get
Prove it	Each company knows you by its own private email address. TraceErase watches those addresses, reads every reply with AI, and records everything in a tamper-proof ledger	A clear status for every company (Verified, Partly deleted, Ignored, Leaked), a deletion certificate, and a ready-made complaint with evidence if anyone breaks the law

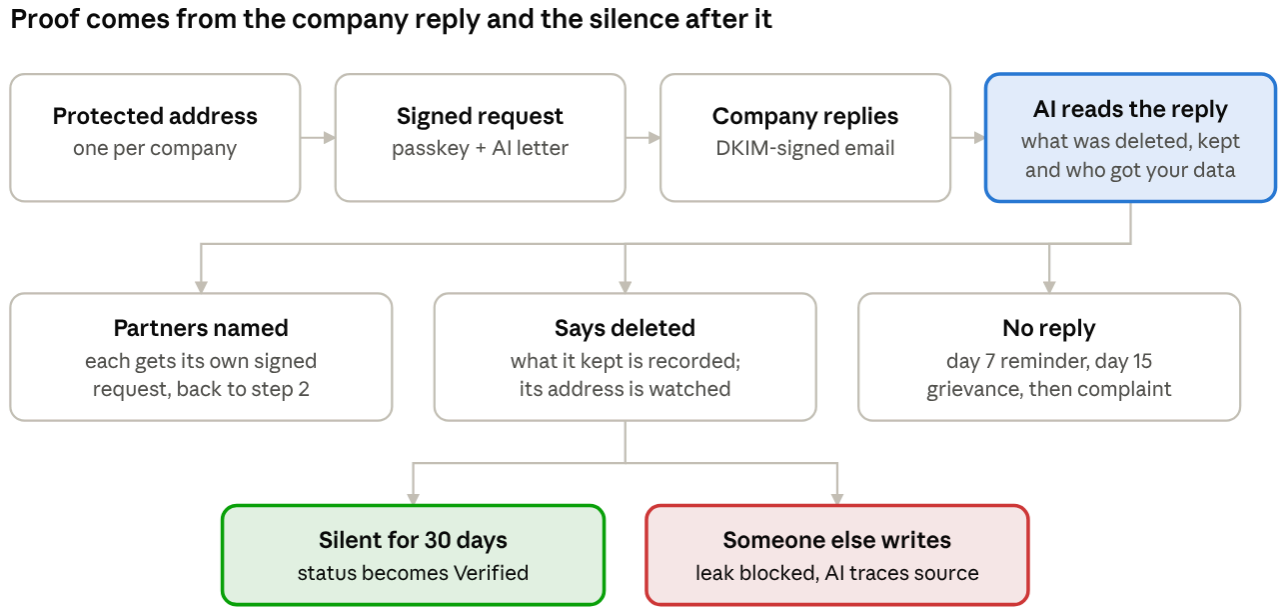
The big idea: a unique identity for every company

The smartest part of TraceErase is simple. Instead of giving every app your real email and phone number, TraceErase gives you a **different protected email address for each company**, plus one **protected phone number**. Messages to them are forwarded to your real inbox and phone, so nothing changes for you in daily use.

But now every message carries a fingerprint:

- Only Zomato knows `zomato.k7p2@te.mail`. If anyone else writes to that address, Zomato (or someone Zomato gave it to) passed it on. **The leak is traced the moment it happens.**
- After Zomato says "we deleted your data", nobody should ever use that address again. If it stays silent, that is real, observable evidence that the data is no longer being used. **Silence becomes proof.**

This turns the question "did they really delete it?" from a matter of trust into something you can watch and measure.



how a deletion becomes proven

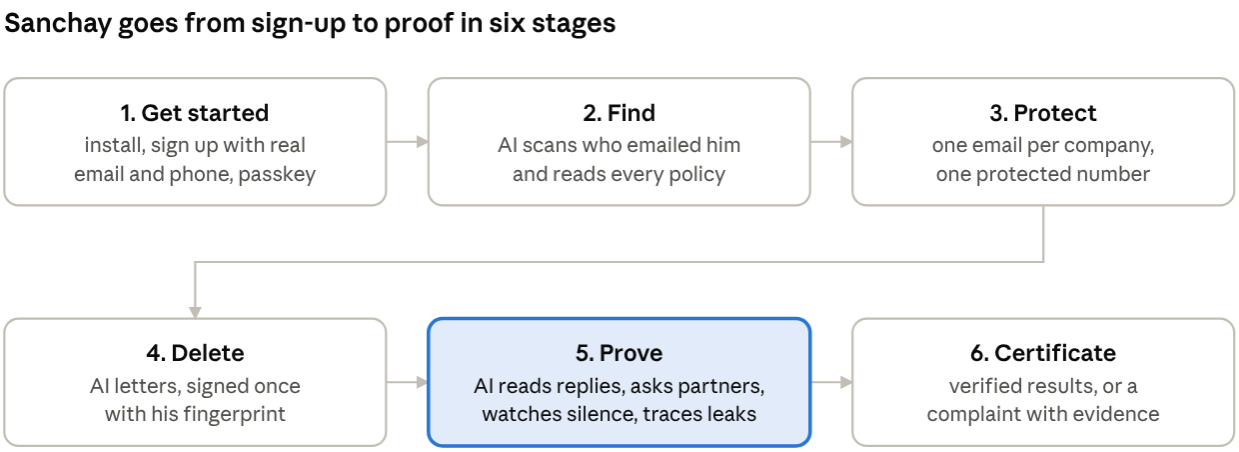
Every reply leads somewhere: partners get their own requests, silence leads to Verified, any stranger's message becomes a traced leak, and no reply leads to a complaint.

The building blocks

1. **Protected identity**: one private email per company and one protected number, forwarding to you.
2. **AI discovery**: finds the companies and reads their privacy policies for you.
3. **AI-written legal requests**: correct DPDP Act letters, in the company's accepted format and in 12 languages.
4. **Passkey signature**: you approve each request with your fingerprint or face, so it is provably yours.
5. **AI reply checker**: reads each company's answer and pulls out what was deleted, what was kept and why, and **which partners got your data**.
6. **Partner chain**: every partner named gets its own deletion request automatically.
7. **Silence verification**: 30 days of silence on a company's address after deletion turns "they said so" into "Verified".
8. **Leak tracing**: a message from a stranger to a protected address is blocked, and AI works out who leaked it.
9. **Automatic escalation**: reminders, a formal grievance, then a complaint to the Data Protection Board with all the evidence attached.
10. **Tamper-proof ledger on blockchain**: every request, reply and event gets a fingerprinted receipt; fingerprints are bundled and anchored on a public blockchain so nobody, not even TraceErase, can quietly change history.
11. **Deletion certificate and public scores**: a certificate anyone can verify, and a public scoreboard of which companies actually delete on time.

How it works: Sanchay's story

Sanchay is a college student in Chennai. Over the years he has signed up for food apps, shopping sites, travel sites, payment apps and games. Lately he gets spam emails and loan calls from companies he has never heard of. He decides to take control. Here is everything that happens, step by step.



Sanchay's journey · 6 stages

Parts 1 to 5 below follow these stages in order; Part 6 is what daily life looks like afterwards.

Part 1: Getting started (about 3 minutes)

Step 1: Sanchay installs TraceErase. He opens the TraceErase website on his laptop, and installs it on his phone from the browser so it sits on his home screen like any app. He also adds the small TraceErase browser extension, which will help him when he signs up for new websites.

Step 2: He picks his language. He chooses English (he could pick Tamil, Hindi or 9 other Indian languages). The whole app, and every letter TraceErase writes for him, can be in that language.

Step 3: He creates his account. He enters his name, his **real** email and his **real** mobile number. TraceErase sends a one-time code to each and he types them in. Why real ones? Because TraceErase will forward the messages that matter to them, so they must be correct and truly his.

Step 4: He creates a passkey. Instead of a password, his phone asks for his fingerprint (or face, or screen lock). This creates a *passkey*: a secret key that never leaves his device. From now on he logs in with a touch, and, more importantly, he will **sign every deletion request** with it, so no one can send a request in his name.

Step 5: He gets his protected identity. TraceErase gives Sanchay a **protected phone number** (for example +91 70000 14317) and the ability to create a **protected email address for any company** (for example `zomato.k7p2@te.mail`). Messages to them reach his real inbox and phone, after TraceErase checks them.

Part 2: Finding where his data is

Step 6: He connects his inbox. TraceErase asks for read-only permission to see **who** has emailed him and the subject lines, never the content of his emails. (If he prefers, he can upload an export of his mailbox instead.) It also looks at the list of apps he logged into with "Sign in with Google".

Step 7: The AI scan runs. In under a minute, the AI goes through hundreds of senders and decides which ones are companies that hold his data (and which are just friends, colleagues or newsletters). It finds **39 companies**: Zomato, Swiggy, Amazon, Flipkart, MakeMyTrip, Paytm, Uber and more. For each one it notes the category, what data it probably has (email, phone, address, payment, location) and when he last used it.

Step 8: AI reads every privacy policy for him. For each company, the AI reads its privacy policy and writes it in three plain lines: what it keeps and for how long, who it shares with, and how to ask it for deletion (and the grievance officer's email). Sanchay never reads a 20-page policy again.

Step 9: He sees the full picture. His dashboard shows a **privacy score**, an AI summary ("You haven't used 14 of these in over a year. Swiggy and MakeMyTrip share with the most partners. Start with those."), and a **data map**: a picture of who has his data and who they passed it on to.

Part 3: Protecting himself from now on

Step 10: New sign-ups use protected details. Next week Sanchay signs up for a new app. The browser extension pops up, fills in a fresh protected email made only for that app plus his protected number, and shows an AI warning from the policy: "Shares your data with advertising partners. Keeps it for 5 years." He decides with full knowledge.

Step 11: Old accounts are moved over. For apps where he used his real email, TraceErase guides him in four steps: change the email in that app to a protected one; send an AI-written request asking the company what data it holds about him; let AI read the answer; decide to keep the account or delete it. Now even his old accounts carry a unique fingerprint.

Step 12: His protected number filters calls and SMS. Delivery updates and OTPs from apps he uses ring through. Loan agents, lottery scams and unknown marketers are blocked by AI, and logged so he can see what was stopped.

Part 4: Deleting his data

Step 13: He chooses what to delete. He taps **Delete my data** and selects Zomato and MakeMyTrip (plus a few apps he no longer uses). TraceErase adds any partners already known to have his data. The summary shows how each request will go: by email to the

grievance officer, or through the company's web form.

Step 14: AI writes the letters. For each company, the AI writes a formal deletion request under the DPDP Act: who he is (by the identifier the company knows), what he wants erased, that partners must erase it too, and that he expects confirmation with a reference number. Sanchay can read and edit every letter.

Step 15: He signs once with his fingerprint. He taps **Sign and send** and touches the fingerprint sensor. His passkey signs the exact content of the request (request number DR-0001). This signature proves the request came from him and was not changed later. A receipt of the signature is recorded in the ledger.

Step 16: TraceErase sends everything. Each letter goes from the company's own protected address to the company's grievance officer, so the reply comes back to TraceErase. Where a company only accepts a web form, TraceErase prepares the text and opens the form for him to paste and submit. If a company replies with a verification code ("Enter this OTP to confirm your deletion"), TraceErase fills it automatically if he allowed it.

Part 5: Proving what really happened

Step 17: Day 1, Zomato replies. The reply arrives at Sanchay's Zomato address. The AI reply checker reads it and reports:

- **Genuine?** Yes: the email is digitally signed by zomato.com (DKIM) and has a reference number.
- **Deleted:** account profile, saved addresses, order history.
- **Kept:** tax invoices, because GST law requires invoices to be kept for 6 years.
- **Shared with:** Razorpay (payments) and AdNexa (advertising).

Zomato's status becomes **Partly deleted (kept by law)**, and a receipt with the details is written to the ledger.

Step 18: The partners are asked too, automatically. Within seconds, TraceErase sends deletion requests to Razorpay and AdNexa, quoting Zomato's own written statement that it shared Sanchay's data with them. Sanchay did nothing; he just gets an alert: "Zomato shared your data with Razorpay and AdNexa. We asked them to delete it too."

Step 19: Day 3, Razorpay replies. It deleted his contact details and saved cards, and kept payment transaction records for 5 years under the Prevention of Money Laundering Act. Status: **Partly deleted (kept by law).**

Step 20: Day 7, the reminders. MakeMyTrip and AdNexa have not replied. TraceErase sends each a polite reminder. MakeMyTrip answers the next day: "Your account and data have been permanently deleted." Status: **Says deleted**, and TraceErase starts watching

Sanchay's MakeMyTrip address.

Step 21: Day 15, AdNexa is still silent. TraceErase sends a formal grievance to AdNexa's grievance officer, as the law requires before going to the regulator. It also prepares an **evidence pack** (Zomato's statement that it shared his data with AdNexa, his signed request, the reminder, the grievance, all with dates and fingerprints) and an AI-drafted **complaint** to the Data Protection Board. If AdNexa's legal time to respond runs out, the complaint is filed with one tap, or automatically if Sanchay turned that on.

Step 22: Day 20, a leak is caught. PizzaPoint, a company Sanchay never used, emails his **Zomato-only** address with "Flat 50% off your first order!" TraceErase knows instantly that this address was given only to Zomato, so:

1. The email is **blocked** before it reaches him.
2. The AI **leak investigator** looks at who had that address: Zomato, Razorpay and AdNexa (as Zomato said). Razorpay confirmed deletion; AdNexa ignored every request for 20 days. Verdict: **most likely leaked through AdNexa**, with the reasons written out.
3. Sanchay is alerted, with options: send PizzaPoint its own deletion request, give Zomato a fresh address (the old one stops working), and **add the leak to the complaint** against AdNexa. He taps all three.

Step 23: Day 38, MakeMyTrip is Verified. Nobody has written to Sanchay's MakeMyTrip-only address for 30 days since it said "deleted". The status changes from Says deleted to **Verified**: the company said it deleted the data, and the evidence agrees.

Step 24: He gets his deletion certificate. TraceErase issues a **Deletion Certificate** for request DR-0001: one row per company with the result (Verified, Partly deleted and what was kept and under which law, or Ignored with the complaint reference), and a QR code. Anyone who scans it can check it against the ledger and see it has not been altered. He can save it as a PDF and keep it.

Part 6: Everyday life with TraceErase

- **Ask TraceErase:** he can ask the assistant, by typing or by voice, "How did PizzaPoint get my email?" or "Which apps should I delete next?" and get answers drawn only from his own records.
- **Daily summary:** a two-line update each day ("Razorpay confirmed deletion. AdNexa's complaint is ready to file.").
- **Activity log:** every email, SMS and call to his protected identity, and whether it was forwarded or blocked, and why.
- **Alerts:** for replies, leaks, deadlines and complaints that need him.
- **Control:** he can rotate any protected address, export all his data, or delete his TraceErase account (and TraceErase gives him a receipt for that too).

Inside each step: how the technology works

This section explains the machinery behind Sanchay's story, one part at a time, in plain words.

1. The protected identity

- **One email per company.** Each address looks like `companyname.randomcode@te.mail`. The random code makes it impossible to guess. TraceErase keeps a private record of which address belongs to which company.
- **Allowed senders.** Each address has a list of who may write to it: at first only that company's own domain (for example zomato.com). If the company names a legitimate partner (like its payment company), that partner can be allowed too. Everyone else is blocked.
- **Forwarding.** Allowed messages are forwarded to Sanchay's real inbox, marked with which protected address they came through. Order updates, receipts and OTPs work exactly as before.
- **Spoof check.** Every real company email carries a digital signature called DKIM. If an email *claims* to be from zomato.com but the signature fails, it is a fake (phishing) and is blocked.
- **Rotation.** If an address leaks, Sanchay can replace it with a new one in one tap. The old one stops working, so the leaker's copy becomes useless.
- **The protected number** works the same way for SMS and calls, with AI deciding which messages are genuine (OTP, delivery, bank) and which are spam or scams.

2. AI discovery

TraceErase reads only the **sender and subject** of emails (never the body), plus the list of apps used with "Sign in with Google". The AI looks at each sender domain and decides: is this a company that holds his data, and what kind of data? A rule-based list of well-known companies backs up the AI, so the scan still works if the AI is unavailable.

3. AI policy reader

For each company, TraceErase fetches its privacy policy page and the AI extracts: what is kept and for how long, which kinds of partners receive data, how deletion requests are accepted (email, web form, in-app), and the grievance officer's email. This tells TraceErase **where** and **how** to send each request, and tells Sanchay what to expect (for example, "invoices will be kept by law").

4. AI request writer

The AI writes each letter in the right legal form: it cites Sanchay's right to erasure under the DPDP Act, names the identifier the company knows him by, asks for erasure by the company **and its processors**, asks for a reference number and a list of anything kept with the legal reason, and states the deadline. It writes different letters for different stages: deletion request, data access request, reminder, formal grievance, and a partner request that quotes the original company's disclosure. Letters can be written in 12 languages.

5. The passkey signature

Before sending, TraceErase calculates a **fingerprint** (a SHA-256 hash) of the exact request: number, date, companies, wording. Sanchay's passkey signs that fingerprint when he touches the sensor. Anyone can later check that the signature matches the request, which proves two things: Sanchay approved it, and nobody changed it afterwards.

6. The AI reply checker

Every reply to a request lands at a protected address, so TraceErase sees it first. The AI reads it strictly and answers:

- Is this a genuine confirmation, a refusal, or just an acknowledgement ("we received your request")?
- Is it digitally signed by the company's domain (DKIM)? Does it give a reference number?
- What exactly was deleted?
- What was kept, and under which law?
- **Which other companies received the data?**

The answers are saved as a receipt, and the status updates automatically.

7. The partner chain

When a reply names partners, TraceErase records a **disclosure receipt** ("Zomato stated on 23 Oct that it shared email, phone and payment details with Razorpay for payment processing"), adds each partner to Sanchay's list, and sends each one its own deletion request. If a partner names further partners, the chain continues. Under the DPDP Act, a company must make its processors erase the data too, so the company's own statement becomes evidence if a partner ignores the request.

8. Silence verification

After a company says "deleted", TraceErase watches the address that **only that company** knew. If no one uses it for **30 days**, the status becomes **Verified**. The logic: a company that still holds and uses the data (for marketing, sharing or selling) would cause messages to that address. Thirty days of silence is real, measurable evidence that the data is no longer in use. Any message during that time is treated as a warning or a leak.

9. Leak tracing

When a stranger writes to a protected address:

1. The message is blocked and saved as evidence (sender, time, subject, fingerprint).
2. The AI leak investigator lists everyone who legitimately had that address: the company it was made for, plus every partner that company named in writing.
3. It weighs the evidence: who confirmed deletion, who ignored requests and for how long, whose purpose was advertising or marketing.
4. It names the most likely source with a confidence level and its reasons, and shows the chain: *You → Zomato → AdNexa → PizzaPoint*.

10. Automatic escalation

When	What TraceErase does
Day 0	Signed request sent
Day 7	Reminder if there is no reply
Day 15	Formal grievance to the company's grievance officer; evidence pack and complaint prepared
When the legal time runs out (up to 90 days under the DPDP Rules)	Complaint filed with the Data Protection Board, with one tap or automatically

The DPDP Act asks people to go through the company's grievance process before approaching the Board, so TraceErase follows that order and keeps the proof of each step.

11. The tamper-proof ledger and blockchain

Every important event becomes a **receipt**: the signed request, each reply, each disclosure, each reminder, each leak, each investigation, each certificate. Each receipt gets a SHA-256 fingerprint. Every minute, new fingerprints are bundled into a **Merkle tree** (a

structure that combines many fingerprints into one "root" fingerprint), and each root is chained to the one before it. The roots are then **anchored on a public blockchain** (Polygon), which nobody can edit.

What this gives: if anyone (a company, an attacker, even TraceErase itself) changes a single letter in an old receipt, its fingerprint no longer matches the anchored root, and verification fails for everyone to see. Only fingerprints go on the blockchain, **never personal data**.

12. Certificate, evidence pack and complaint

- **Deletion certificate:** a summary of one request with every company's result, the fingerprint of each supporting receipt, and a QR code linking to a public verification page.
- **Evidence pack:** the receipts that prove a case (how the company got the data, the signed request, reminders, grievance, leak messages, the AI investigation), each with time and fingerprint.
- **Complaint:** an AI-drafted complaint to the Data Protection Board, built only from the evidence pack, which Sanchay reviews before filing.

What every status means

Each company in Sanchay's list carries one status, so he can see the truth at a glance. Strongest proof is at the top.

Status	What it means	What it is based on	What TraceErase does next
Verified	Deleted, and the evidence agrees	The company confirmed in a signed email, and its private address stayed silent for 30 days afterwards	Keeps watching; any later message becomes a leak alert
Partly deleted (kept by law)	Deleted, except some records the law requires it to keep	Signed confirmation that names what was kept and the law (for example tax invoices under GST law)	Records what was kept and why; it appears on the certificate
Says deleted	The company claims deletion; being checked	Signed confirmation, silence period still running	Counts down the 30 days

Status	What it means	What it is based on	What TraceErase does next
Waiting	Request sent, no answer yet	Delivery record of the signed request	Reminder on day 7, grievance on day 15
No response	Reminder sent and still ignored	Delivery records of the request and reminders	Grievance, evidence pack, complaint
Overdue	Ignored past the deadline	Delivery records of the request, reminder and grievance	Complaint to the Data Protection Board
Refused	The company said no	The company's own reply	AI explains whether the refusal is lawful; complaint if not
Leak	Someone who should not have the data used it	A blocked message to a protected address	Investigation, address rotation, evidence added to the complaint

"Kept by law" is not a failure. Some laws require companies to keep certain records, such as tax invoices for about 6 years or payment records for 5 years. TraceErase makes sure the company says **exactly what** it kept and **under which law**, so Sanchay knows, and anything kept without a legal reason can be challenged.

The AI agents

AI is not a decoration in TraceErase; it does the jobs a person cannot do at scale: reading hundreds of senders, dozens of long policies, and every company's reply, then reasoning about evidence. TraceErase uses **Google Gemini** for these 12 agents.

#	AI agent	What it does	Why it needs AI
1	Discovery agent	Finds which email senders are companies holding your data, with category and data types	Sender names are messy; telling a company from a friend or a newsletter needs judgement
2	Scan summariser	Turns the scan into one useful insight and where to start	Prioritising needs reasoning over many facts at once

#	AI agent	What it does	Why it needs AI
3	Policy reader	Reads each privacy policy: what is kept, how long, who it is shared with, how to delete, grievance email	Policies are long legal text written differently by every company
4	Request writer	Writes deletion, access, reminder, grievance and partner letters under the DPDP Act, in 12 languages	Each letter must fit the company, the stage and the law
5	Reply checker	Decides if a reply truly confirms deletion; extracts what was deleted, what was kept and why, and which partners got the data	Replies are free text; spotting a vague "we've processed your request" versus a real confirmation needs careful reading
6	Leak investigator	Works out who most likely passed a protected address to a stranger, with reasons and confidence	Weighing several pieces of evidence, like a detective
7	Complaint drafter	Writes the complaint to the Data Protection Board from the evidence pack	Turning evidence into a clear, formal legal complaint
8	SMS and call guard	Allows genuine messages (OTP, delivery, bank) and blocks spam and scams on the protected number	Scam messages change wording constantly
9	Ask TraceErase assistant	Answers questions by text or voice, using only Sanchay's own records	Natural conversation, grounded in his data
10	Daily summary	Two plain sentences on what changed today	Summarising many events briefly
11	Regulator pattern finder	Spots repeat offenders across many complaints (for the regulator view)	Patterns across many cases
12	Interface translator	Translates the whole app into the chosen language	Natural, accurate translation of the interface

Privacy rule for all AI: before anything is sent to the AI, Sanchay's name, email and phone number are replaced with placeholders like [NAME] and [EMAIL]. The AI never sees who he is.

Safety net: every agent has a rule-based backup. If the AI is unavailable, TraceErase keeps working with simpler rules and clearly marks those results, so nothing stops.

Keeping Sanchay's own data safe

A privacy tool must be private itself. TraceErase is designed to hold as little as possible and to protect what it holds.

Protection	How it works
No passwords	Login and signing use passkeys (fingerprint, face or screen lock). The secret key never leaves his device, so there is nothing to steal or phish.
Email content stays private	The inbox scan reads only senders and subjects, with read-only permission he can withdraw any time.
AI never sees who he is	Name, email and phone are masked before any AI call.
Verification codes are protected	One-time codes are stored only as scrambled hashes, and expire quickly.
Secure sessions	Logins use signed, httpOnly cookies that web pages cannot read.
Nothing personal on the blockchain	Only fingerprints (hashes) are anchored; a fingerprint cannot be turned back into the data.
Phishing blocked	Emails pretending to be a company (failed DKIM signature) are stopped.
He stays in control	He can rotate any address, export everything TraceErase holds about him, or delete his account; TraceErase gives him a receipt for its own deletion too.
Nothing sent without approval	Deletion requests go out only after he signs them with his passkey.

What companies and the regulator see

TraceErase is built for people, but it changes how companies behave and gives the regulator better cases.

Companies

- **Nothing to install, nothing to join.** Companies receive normal, polite, legally correct emails at their grievance address, exactly what the law already asks them to handle.
- **Clear requests.** Each request names the identifier, the law, what is asked and the deadline, so a company that wants to comply can do it quickly.
- **A public score.** TraceErase publishes a **transparency board**: for each company, how often it deleted on time, average days to respond, leaks traced to it, and a grade from A+ to F. Companies that delete properly look good in public; companies that ignore people or leak data are visible.

The Data Protection Board (regulator)

- **Complaints with evidence.** Every complaint arrives with a fingerprinted evidence pack: the signed request, delivery records, reminders, the grievance, disclosures and leaks. The Board can check every item against the blockchain.
- **Patterns, not single cases.** An AI view groups complaints and highlights repeat offenders (for example, "AdNexa: ignored deletions in many complaints and linked to traced leaks"), so the Board can act where it matters most.

Questions people ask

About proof

How can you prove a company deleted my data when you can't see inside its servers? We combine three kinds of evidence that the company cannot fake: (1) its own **digitally signed reply** saying what it deleted and kept; (2) **silence** on the address only it knew, for 30 days after; and (3) a **tamper-proof record** of every step on the blockchain. Together they turn "trust us" into evidence. If any of them breaks (no reply, a vague reply, or a message on the address), the status shows it.

What if the company lies, says "deleted", and keeps using my data? Then it will show up. If it emails the address, shares it or sells it, a message arrives at that address. TraceErase blocks it, marks it as a leak, and the company's own signed "deleted" email becomes evidence that it lied.

What if a company keeps my data quietly and never uses it? Silence cannot see data sitting unused in a backup. No outside tool can. That is why the status says "Verified" only when both the signed confirmation and the silence agree, and why the confirmation must list anything kept. If the company is caught later, its signed statement is on record. This is the strongest proof possible from outside a company.

Why 30 days of silence? A company or partner that still uses data for marketing or shares it usually acts within weeks. Thirty days is long enough to catch that, short enough to give a clear answer. Watching does not stop after 30 days: any later message still raises a leak alert.

Why use blockchain instead of a normal database? A normal database can be edited by whoever runs it, including us. By anchoring the fingerprints of all receipts on a public blockchain, nobody can change the history without it being detected. That is what makes the records usable as evidence.

Is my personal data put on the blockchain? No. Only fingerprints (hashes) go there. A fingerprint proves a record has not changed, but cannot be turned back into the record.

About companies and partners

How do you find out who a company shared my data with? In two ways. First, the deletion request asks the company to list everyone it shared the data with, as the law makes it responsible for its processors. The AI reads the reply and extracts the names. Second, the AI policy reader shows what kinds of partners the company says it uses, before you even ask.

What if the company doesn't name its partners? The policy still tells us the types of partners. And if your data later turns up somewhere else, the protected address proves it came through that company, whether it named the partner or not.

What if a partner leaks my data, not the company itself? The leak investigator looks at everyone who legitimately had the address and the evidence about each. In Sanchay's case it pointed to AdNexa, the ad partner that ignored his request, not Zomato.

What if a company never replies? Reminder at day 7, formal grievance at day 15, then a complaint with the full evidence pack to the Data Protection Board when the legal response time runs out. Silence from a company is recorded as evidence too.

What if the company keeps some data because of a law? That can be legal (for example tax invoices or payment records). The status becomes "Partly deleted (kept by law)", and the certificate shows exactly what was kept and under which law.

What if a company asks me to verify who I am? If it sends a one-time code to the protected address, TraceErase can fill it in automatically (only if Sanchay allows it). If it needs something else, TraceErase alerts him.

What about companies outside India? The request writer uses the law that applies: the DPDP Act in India, the GDPR's right to erasure in Europe, or similar laws elsewhere. The proof method is the same everywhere.

About using TraceErase

Does TraceErase read my emails? No. With your permission it reads only who sent each email and the subject line, never the content, and you can withdraw access at any time. It does read replies to your own deletion requests, which arrive at your protected addresses.

Will I still get OTPs, order updates and delivery messages? Yes. Messages from the company you signed up with are forwarded to your real inbox and phone as normal.

What about old accounts where I used my real email? TraceErase helps you switch them to a protected address first. From then on they are traceable like any new account.

Can someone else delete my accounts by pretending to be me? No. Every request must be signed with your passkey, which needs your fingerprint or face on your device.

What if I lose my phone? Passkeys are usually backed up by your phone's password manager (iCloud Keychain or Google Password Manager), and you can register more than one device. You can also log in with codes sent to your verified email and phone.

Can a spammer just guess my protected address? Practically no. Each address has a random code, and messages that fake a company's identity fail the DKIM check and are blocked.

Is it legal for TraceErase to send requests for me? The right to erasure is yours under the DPDP Act. Each request is your own, signed by you, and TraceErase is the tool you use to write, send and track it.

What does TraceErase keep about me, and can I delete it? Only what it needs to work: your account details, protected addresses, requests and receipts. You can export it all or delete your account at any time, and you get a receipt for that deletion too.

How is this different from emailing companies myself? You would have to find every company, find its grievance address, write each letter, track deadlines, and you would still only have the company's word. TraceErase does all of that, also covers the partners, and adds the evidence that shows whether the deletion really happened.

Why TraceErase is different

Most privacy tools stop at sending a request and showing the company's reply. TraceErase goes further in five ways:

1. **It proves, not just asks.** Statuses are based on evidence (signed replies, silence, leaks), not on the company's word alone.
2. **It follows the data to the partners.** Every partner a company names is asked too, automatically, and the company's own statement is kept as evidence.
3. **It catches leaks at the source.** One address per company means every leak points to who leaked it.
4. **Its records can't be altered.** Every step is fingerprinted and anchored on a blockchain, so the evidence holds up in a complaint.
5. **It works today, with every company.** Nothing for companies to install or join; it uses email, the law and AI.

Who benefits

Who	What they get
People like Sanchay	Control over their data, fewer spam calls and emails, and real proof
Honest companies	Clear requests and a good public score for deleting on time
The Data Protection Board	Complaints that come with verifiable evidence, and a view of repeat offenders
Society	Pressure on companies to share less and delete properly

Honest limits

We say these openly, because they make the solution more credible, not less:

- **Silence is strong evidence, not a look inside the servers.** No outside tool can inspect a company's databases or backups; Verified means the company confirmed in writing and nobody used the data afterwards.
- **Protected identity works best from the start.** Old accounts using a real email need to be switched to a protected address first.
- **Some data may lawfully stay.** TraceErase makes sure it is named and justified, but cannot force deletion of what the law requires a company to keep.
- **Filing with the Board follows the law's order.** The company's grievance process comes first, and the DPDP Rules give companies up to 90 days to respond, so some cases take time.

- **Partner discovery depends partly on the company's answer.** If a company hides a partner, the leak tracing still catches misuse, but only when the data is actually used.

Glossary

Term	Simple meaning
Data Principal	The person the data is about (Sanchay)
Data Fiduciary	The company that decides how the data is used (Zomato)
Data Processor	A company that handles data for the fiduciary (Razorpay, an ad network)
DPDP Act, 2023	India's Digital Personal Data Protection Act, which gives people rights over their data, including erasure
DPDP Rules, 2025	The rules that put the Act into practice, including the 90-day response limit
Data Protection Board	India's digital regulator that handles complaints under the DPDP Act
Grievance officer	The person a company must name to handle privacy complaints
Protected email / number	A private address or number made by TraceErase, forwarding to your real one
Passkey	A login and signing key protected by your fingerprint, face or screen lock; it never leaves your device
DKIM	A digital signature on emails that proves which domain really sent them
Hash / fingerprint (SHA-256)	A short code calculated from a record; change one letter and the code changes completely
Merkle tree	A way to combine many fingerprints into one root fingerprint, so any change can be detected
Blockchain anchoring	Writing that root fingerprint to a public blockchain (Polygon) so nobody can secretly change history

Term	Simple meaning
Receipt	A fingerprinted record of one event (request, reply, disclosure, leak)
Disclosure	A company's written statement of who it shared your data with
Silence verification	Watching a company-only address for 30 days after deletion; no messages means the data is not being used
Leak	A message to a protected address from someone who should not have it
Evidence pack	The set of receipts that prove a case
Deletion certificate	A verifiable summary of a deletion request's results, with a QR code
Gemini	Google's AI model, which powers TraceErase's AI agents

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